Krylov project writeup

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1 Abstract

The aim of this project was to investigate how quantum spin chains transform from integrable to chaotic using Krylov Complexity as a probe for chaotic behaviour. In particular, we looked at the Heisenberg XXZ spin chain model, with Next-to-Nearest-Neighbor (NNN) interactions to break integrability, with coupling $g \in (0, 1)$. We looked at chains from length 9 to 16, constructed the Krylov basis using the Lanczos algorithm, and then computed normalized Krylov complexity over time. The chaotic systems, $g = 1,\ g = 0.5$, tend to reach higher plateau values in Krylov complexity. Additionally, we have shown that chaotic systems reach their respective plateaus faster than integrable ones. To compliment our Krylov complexity findings, we also looked into Shannon entropy of eigenstates. Shannon entropy shows a functional dependence on energy in chaotic systems, $g = 1$. The appearance of this dependence becomes more extreme with longer length chains. A key result in terms of the transition from integrability to chaos, is that as the chain length approaches the thermodynamic limit, we see a clearer distinction between integrable and chaotic systems. The main takeaway from our research is that K-complexity proves to be a successful tool in determining if a spin chain is chaotic or integrable, with clearer transitions emerging as chain length increases toward the thermodynamic limit.

2 Introduction

Quantum many-body systems can often be classified as integrable or chaotic. This information is crucial in understanding thermalisation, and the quantum-to-classical transition. Traditionally, quantum chaos has been studied using spectral statistics. Integrable systems show a Poissonian level spacing distribution, while chaotic systems exhibit Wigner-Dyson statistics with level repulsion. A drawback of statistical analysis is that it only provides us with static information about the systems in question. We do not obtain details on how they evolve over time.

Krylov complexity has been introduced as a new tool in the investigation of quantum chaos. It probes information propagation directly, and provides a dynamical view of quantum chaos rather than a static one. K-complexity measures how quantum states "spread" in Hilbert space over time, more specifically, it measures how far a state spreads in a Krylov subspace. Krylov subspaces are generated by repeated application of the Hamiltonian and form an orthonomal basis representation of accessible Hilbert space.

Quantum spin chains are many body systems ideal for studying integrable-to-chaotic transitions. The Heisenberg XXZ spin chain model is a well known integrable system. In our project, we induce chaos on this model by adding an NNN interaction, with a coupling coefficient such that we can vary the interaction strength. We look at the transition from integrable to chaotic in multiple contexts, by varying the strength of the NNN interaction and the size of the Hilbert space.

3 Background

3.1 Krylov-Complexity

The order r Krylov subspace is generated by an nxn matrix A and a vector b of dimension n,

$$K_r(A, b) = span\{b, Ab, A^2b, ..., A^{r-1}b\} = span\{A^r b\}_{r\geq 0} \tag{3.1}$$

Krylov complexity measures how much that vector "spreads" in Krylov space. For our purposes, K-complexity is how much an operator or state spreads in Krylov space over time as they evolve.

K-complexity has been demonstrated in previous works to be a useful tool in determining whether a system is integrable or chaotic [1]. Computing this value requires several steps, constructing an orthonormal basis for the Krylov space, time evolving our state, and finally computing the complexity for said state. In the following sections we will discuss how that is accomplished.

3.2 Lanczos Algorithm

The Lanczos algorithm is an iterative method, which constructs an orthonormal basis for the Krylov space $\{|K\rangle\}_{n=0}^{K-1}$. To begin, we take a Hermitian operator H acting on a D-dimensional Hilbert space, from which we take the probe vector $|\Omega\rangle$.

The Krylov space associated with $|\Omega\rangle$ is given by,

$$\mathcal{H}_{|\Omega\rangle} := span\{H^k\,|\Omega\rangle\}_{k\geq 0}, \tag{3.2}$$

where K is the Krylov dimension, and is given by the number of eigenspaces of H with which $|\Omega\rangle$ overlaps, thus $1 \leq K \leq D$.

The Krylov basis, $\{|K_n\rangle\}_{n=0}^{K-1}$, is generated by iteratively acting in each step with H over the previously constructed basis vector, and subtracting from it a certain linear combination of the preceding basis elements, while demanding that the next basis vector has a minimal norm. Thus the eigenvalues of H contained in the Krylov space alternate dominance in the different basis elements, which ensures that no eigenvalues lead over the rest. As seen in [4], the Lanczos algorithm is a more refined version of the Gram-Schmidt orthogonalisation process in which we get two series of coefficients, $\{a_n\}_{n=0}^{K-1}$, and $\{b_n\}_{n=1}^{K-1}$, which together with $\{|K_n\rangle\}_{n=0}^{K-1}$ suffice for us to solve for the full resolution of the eigenvalue problem restricted to the eigenspaces of H that $|\Omega\rangle$ probes.

The algorithm itself consists of first assigning the initial basis element to be the probe vector, $|\Omega\rangle$, we assume, for simplicity, that $|\Omega\rangle$ is normalised to unity, this gives,

$$|K_0\rangle = |\Omega\rangle\,. \tag{3.3}$$

Now we apply H to this basis element and subtract from it a vector of magnitude x in the direction of $|K_0\rangle$, which creates a non-normalised version of the next basis element.

$$|A_1\rangle = H\,|K_0\rangle - x\,|K_0\rangle\,, \tag{3.4}$$

where x is some number determined from the fact that $|A_1\rangle$ has a minimal norm, thus we get,

$$G(x) = ||\,|A_1\rangle\,||^2 = ||H\,|K_0\rangle\,||^2 + x^2 - 2x\,\langle K_0|\,H\,|K_0\rangle\,, \tag{3.5}$$

we can see that $G(x)$ is a convex parabola, thus attaining a minimum when,

$$x = \langle K_0|\,H\,|K_0\rangle = a_0, \tag{3.6}$$

a_0 being the first term of the first of two sequences of Lanczos coefficients.

As $|A_1\rangle$ is orthogonal to $|K_0\rangle$, we have that

$$\langle K_0|\,A_1\rangle = \langle K_0|\,(H\,|K_0\rangle - \langle K_0|\,H\,|K_0\rangle) = 0. \tag{3.7}$$

Now to normalise $|A_1\rangle$, we define

$$b_1 = ||\,|A_1\rangle\,|| = \sqrt{G(a_1)}, \tag{3.8}$$

thus we find the second Krylov basis element to be

$$|K_1\rangle = \frac{1}{b_1}\,|A_1\rangle = \frac{1}{b_1}(H - a_0)\,|K_0\rangle\,. \tag{3.9}$$

The second basis element is found by a similar procedure, this time applying H to $|K_1\rangle$, and subtracting a linear combination of vectors in the directions of $|K_1\rangle$, and $|K_2\rangle$, with magnitudes x and y respectively. Thus we have,

$$\begin{aligned} |A_2\rangle &= H\,|K_1\rangle - x\,|K_1\rangle - y\,|K_0\rangle\,, \\ G(x,y) &= ||\,|A_2\rangle\,||^2 = ||H\,|K_1\rangle\,||^2 + x^2 - 2x\,\langle K_1|\,H\,|K_1\rangle + y^2 - 2yRe(\langle K_1|\,H\,|K_0\rangle). \end{aligned} \tag{3.10}$$

Due to the fact that the Krylov basis is orthonormal, $G(x,y)$ contains no mixing terms of x and y, thus to minimise the function we simply take the corresponding partial derivatives.

$$\begin{aligned} \partial_y G(x,y) = 0 \implies x &= \langle K_1|\,H\,|K_1\rangle = a_1, \\ \partial_x G(x,y) = 0 \implies y &= Re(\langle K_1|\,H\,|K_0\rangle). \end{aligned} \tag{3.11}$$

the y-value can then be further simplified using our knowledge of the construction of $|K_0\rangle$, and $|K_1\rangle$, thus giving us,

$$\langle K_1|\,H\,|K_0\rangle = \langle K_1|\,(b_1\,|K_1\rangle + a_0\,|K_0\rangle) = b_1, \tag{3.12}$$

We find the minimum at $(x,y) = (a_1, b_1)$.

We again impose orthonormality of $|K_0\rangle\,, |K_1\rangle$ to $|A_2\rangle$ and get,

$$\begin{aligned} \langle K_0|A_2\rangle &= \langle K_0|(H - a_1)|K_1\rangle - b_1\,|K_0\rangle = 0, \\ \langle K_1|A_2\rangle &= \langle K_1|(H - a_1)|K_1\rangle - b_1\,\langle K_1|K_0\rangle = 0. \end{aligned} \tag{3.13}$$

we again normalise $|A_2\rangle$ to find the next b-coefficient and find the next Krylov basis element,

$$\begin{aligned} b_2 :&= ||\,|A_2\rangle\,|| = \sqrt{G(a_1,b_1)}, \\ |K_2\rangle &= \frac{1}{b_2}\,|A_2\rangle = \frac{1}{b_2}(H - a_1)\,|K_1\rangle - b_1\,|K_0\rangle \end{aligned} \tag{3.14}$$

We inductively apply this procedure, as seen in [3], we find an expression for the non-normalised basis elements,

$$|A_n\rangle = (H - a_{n-1})\,|K_{n-1}\rangle - b_{n-1}\,|K_{n-2}\rangle \tag{3.15}$$

this yields an orthonormal basis for $n \geq 2$, when

$$\begin{aligned} a_n &= \langle K_n|H|K_n\rangle \\ b_n &= \sqrt{\langle A_n|A_n\rangle} \end{aligned} \tag{3.16}$$

We also find that the normalised Krylov elements satisfy,

$$|K_n\rangle = \frac{1}{b_n}\,|A_n\rangle = \frac{1}{b_n}(H - a_n)\,|K_{n-1}\rangle - \frac{b_{n-1}}{b_n}\,|K_{n-2}\rangle\,. \tag{3.17}$$

we must only account for $n \leq K-1$, as for the last step, $n = K$, we have exhausted all directions, and so can only find the null vector, thus we terminate the process, as the last b-coefficient $b_k = 0$ as,

$$|A_K\rangle = 0 \iff b_K = 0. \tag{3.18}$$

3.3 State Complexity

In this project, we focused primarily on state complexity rather than operator complexity. The dimension of the operator space is the square of the state space (the Hilbert space), so performing calculations in the operator space is considerably heavier. However, calculating operator complexity can be done with roughly the same process in operator space. [3]

Firstly, we apply the Lanczos algorithm for a time evolved state for the Schrödinger picture, given by,

$$|\psi(t)\rangle = e^{-itH}\,|\psi\rangle\,, \tag{3.19}$$

where $|\psi\rangle$ is the initial state of the system, H is the hamiltonian and $\hbar$ has been set to 1.

This gives us an orthonormal Krylov basis $\{|K_n\rangle\}_{n=0}^{K-1}$, which generally has two sets of coefficients, $\{a_n\}_{n=0}^{K-1}$, and $\{b_n\}_{n=0}^{K-1}$, related by,

$$b_n\,|K_n\rangle = (H - a_{n-1})\,|K_{n-1}\rangle - b_{n-1}\,|K_{n-2}\rangle\,, \tag{3.20}$$

where $|K_{-1}\rangle \equiv |K_K\rangle \equiv 0$.

The time-evolved state can then be decomposed over the Krylov basis as,

$$|\psi(t)\rangle = \sum_{n=0}^{K-1} \phi_n(t)\,|K_n\rangle \tag{3.21}$$

In the Krylov basis the Hamiltonian is tridiagonal, where the b-coefficients now act as the hopping amplitudes, and the a-coefficients represent the level potential energies on each site of the Krylov chain. This Hamiltonian is given by,

$$H_K = \sum_{n=0}^{K-1} a_n\,|K_n\rangle\,\langle K_n| + \sum_{n=0}^{K-2} b_{n+1}(|K_n\rangle\,\langle K_{n+1}| + |K_{n+1}\rangle\,\langle K_n|). \tag{3.22}$$

The Schrödinger equation dictated by H_K gives us a new relation for the wave function, which now has diagonal terms controlled by the a-coefficients.

$$i\dot{\phi}_n(t) = a_n\phi_n(t) + b_{n+1}\phi_{n+1}(t) + b_n\phi_{n-1}(t), \tag{3.23}$$

with an initial condition $\phi_n(0) = \delta_{n0}$, and boundary conditions $\phi_{-1} \equiv 0 \equiv \phi_K$, due to the fact that the initial state is a zeroth order Krylov element.

As the unitary preserves the norm of the Krylov space wave function, we can define the Krylov complexity and Krylov entropy for the states. The norm being defined as,

$$\sum_{n=0}^{K-1} |\phi_n(t)|^2 = 1. \tag{3.24}$$

Thus we get the respective equations for the Krylov entropy and Krylov complexity. The Krylov complexity, C_k, is defined as the expectation value of position on the Krylov chain. The Krylov entropy, S_k, is the Shannon entropy of the wave function.

$$\begin{aligned} Sh_K &= -\sum_{n=0}^{K-1} |\phi_n(t)|^2 log|\phi_n(t)|^2 \\ C_K &= \sum_{n=0}^{K-1} n|\phi_n(t)|^2 \end{aligned} \tag{3.25}$$

Due to the fact that the Lanczos coefficients are always real by our construction for the Krylov space wave function, we see that complex conjugation is equivalent to setting $t = -t$.

$$\phi_n^\star(t) = \phi_n(-t). \tag{3.26}$$

Thus a useful property of K-complexity becomes clear, that the K-entropy, and K-complexity are even functions of time. We did not make use of this in our work, however it's still an interesting result.

3.4 Level Statistics

The level statistics for our quantum spin chain system are a distribution of energy eigenvalues for said system, which show whether the system is integrable, in which case the distribution follows Poissonian statistics, or chaotic, for which we see Wigner-Dyson statistics.

For the Wigner-Dyson statistics, there is level repulsion, and thus the energy levels do not overlap, which is not the case for Poissonian statistics. Most typically we see the Gaussian Orthogonal Ensemble, GOE, for which the system has time-reversal and rotational symmetry. For these systems, the Hamiltonian can be represented by symmetric and real matrices. We do, however, rarely also see GUE, Gaussian Unitary Ensemble, for systems in which the Hamiltonians are Hermitian, and time-reversal symmetry is broken, and we also see Gaussian Symplectic Ensemble, GSE, in systems where the rotational symmetry is violated, and in which the Hamiltonian is a Symplectic transformation. However, for our project we only had the opportunity to look at GOE systems.

For this project we wished to compare the distributions of different systems, in our case of different length chains, however to do this we needed to unfold their spectra such that the average densities of states were close to each other. We accomplish this by first ordering the spectrum in increasing energy values, discarding roughly 10-15% of the eigenvalues from the edges of the spectrum in total. We then separated the remaining values into smaller sets of eigenvalues, dividing each set by its appropriate mean level spacing given by,

$$s_{mean} = \frac{(s_1 + s_2 + ... + s_N)}{N} = \frac{E_{N+1} - E_1}{N}, \tag{3.27}$$

where $s_i = (E_{i+1} - E_i)$ are the level spacings for all i, E_i are the eigenvalues, and N is the number of level spacings in the given set of eigenvalues.

This normalised spectrum can be plotted as a histogram to see the shape of the distribution, however it is possible to skip the unfolding step by instead calculating the ratios of consecutive level spacings, r_n given by,

$$r_n = \frac{s_n}{s_{n-1}}. \tag{3.28}$$

In [2], it was found that as the next nearest neighbour, the NNN system transitioned from integrable to chaotic, that the system's distribution transitioned from Poissonian to Wigner-Dyson with steps resembling a distribution between Poissonian and Wigner-Dyson distribution.

We did not use this fact, however to study the actual transition from integrability to chaos, or vice versa, the Brody distribution is used, with a parameter β, [4]

$$P_B(s) = (\beta + 1)bse^{-bs^{\beta+1}}, \text{where } b = \Gamma\left(\frac{\beta+2}{\beta+1}\right)^{\beta+1}$$

.

Whenever β is 0, we have a Wigner-Dyson distribution, however when $\beta = 1$, we get a Poissonian distribution.

3.5 Shannon Entropy

The Shannon entropy of a system is a measure of the complexity of the eigenstates of the Hamiltonian. The entropy tends to be lower for integrable systems than for chaotic ones, and is seen as a fuzzy horizontal band. In chaotic systems, the Shannon entropy can take on any of the Gaussian distributions, or the Wigner-Dyson shape, thus the entropy is a function of energy. Additionally, low entropy usually implies that the state is localised, and as it increases, the states become more and more delocalised. The formula for Shannon entropy is given by,[4]

$$Sh = -\sum_{n=0}^{K-1} |\psi_n(t)|^2 \log |\psi_n(t)|^2. \tag{3.29}$$

The $\psi_n(t)$ here represent the eigenstates of the hamiltonian in a given basis. In our work we use the standard basis however, commonly the eigenbasis is chosen instead for more consistent results.

If the eigenbasis, or any other non-standard basis, is to be used we redefine the eigenstates as $|\psi_\alpha\rangle = \sum_n C_\alpha^n |n\rangle$, with $|n\rangle$ being the basis vectors. Leaving us with,

$$Sh_\alpha^n \equiv -\sum_n |C_\alpha^n|^2 \ln |C_\alpha^n|^2$$

.

4 Numerical Results

A numerical investigation was carried out, studying the transition of the Heisenberg XXZ model with Next to Nearest Neighbour (NNN) interaction. In particular, we investigated the effect of the length of the chain on the Krylov complexity in the integrable and chaotic cases, as well as how complexity changes when the coefficient of the NNN interaction is increased.

4.1 Hamiltonians used

A spin chain is a physical system in which particles fixed in space interact with each other based on their individual spins. Here, we are concerned with spin-1/2 particles, so the Hilbert space dimension of each individual particle is 2 and its general state can be written in the z-spin basis, as follows:

$$|\psi\rangle = a_1 |\uparrow\rangle + a_2 |\downarrow\rangle$$

When we combine multiple spin-1/2 particles into a chain, their Hilbert spaces combine and you represent the overall state of the system with a tensor product of the individual states of each spin. This means that the dimension of the total Hilbert space is 2^L, where L is the length of the chain. However, due to the choice of Hamiltonian we will only have to work with a part of that.

For the numerical investigation, the Heisenberg XXZ model was used. The integrable version of the XXZ Hamiltonian is given below.

$$H = \sum_{i=0}^{L-1} S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + J_{zz} S_i^z S_{i+1}^z \tag{4.1}$$

Here, $S_i^{x,y,z}$ are spin operators in the corresponding directions, acting on the i-th particle in the spin chain.

One can trivially show that this Hamiltonian conserves the total spin of the system, meaning that no new spin-ups or downs are generated, the Hamiltonian merely moves them around. This means that for all of the numerical simulations we can work in a definite total spin sector, which effectively decreases the dimension of the Hilbert space from 2^L to ${}_LC_M$, where M is the total number of up spins in the initial state. $J_{zz} \neq 1$ is required to break the isotropy symmetry. For the numerical work below $J_{zz} = 0.5$ was used.

On its own, the XXZ model is integrable. In order to break integrability we introduce a Next to Nearest Neighbour (NNN) interaction, giving a full Hamiltonian that looks like:

$$H = \sum_{i=1}^{L-1} S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + J_{zz} S_i^z S_{i+1}^z + J_{ch} \sum_{i=1}^{L-2} S_i^x S_{i+2}^x + S_i^y S_{i+2}^y \tag{4.2}$$

Here, J_{ch} represents the strength of the NNN interaction. This allows each particle to impact not only it's nearest neighbours but also the ones next to them. It turns out that this change has a qualitative impact on the dynamics of the system, turning it from integrable to chaotic. Showing the effect that this has on the Krylov complexity is the central goal of this section.

4.2 Lanczos Coefficients

Lanczos coefficients were computed according to the Lanczos algorithm as outlined in Equations 3.15 and 3.16. However, since the calculation is being done numerically, for larger L, the algorithm would fail due to numerical error building up, and the resulting vectors would no longer be orthonormal. To combat this, after every step of the Lanczos algorithm, Graham-Schmidt orthogonalisation was applied to ensure that orthogonality is maintained. A plot of the computed coefficients is shown below:

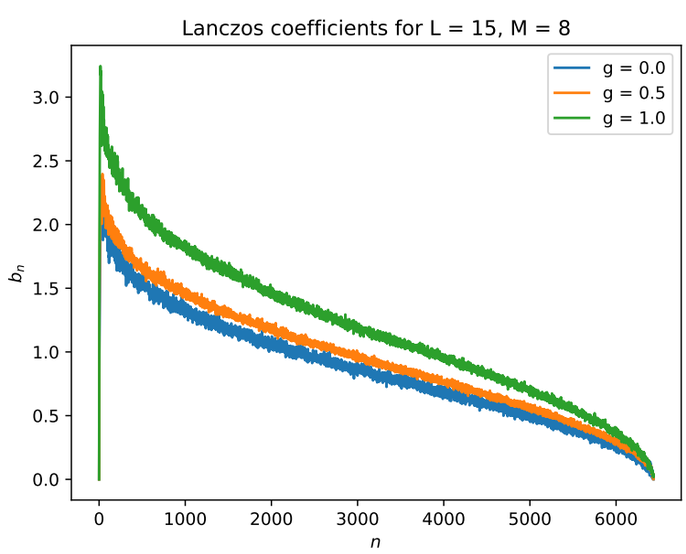


Figure 4.1: Lanczos coefficients

In the above, g takes the place of J_{ch} and represents the strength of the chaotic interaction. The initial state used for the above was

$$\psi_0 = (1, 0, 0, ..., 0)$$

We see that the coefficients follow the expected trend as outlined by E. Rabinovici **and** Sonner [1]. It is also important to note that length and choice of starting state will have a large effect on the Lanczos coefficients. For the state it is important to choose a "poorly mixed state". Since Krylov Complexity (which is linked to the Lanczos coefficients) measures how well a state spreads though the Hilbert space, using a starting state such as:

$$\frac{1}{\sqrt{\dim(\mathscr{H})}}(1, 1, 1, ..., 1)$$

which is already very well spread out, results in the Lanczos coefficients being less smooth, and the Krylov Complexity producing nonsensical conclusions. Similarly, for shorter chains the oscillations are significantly more visible, as seen below.

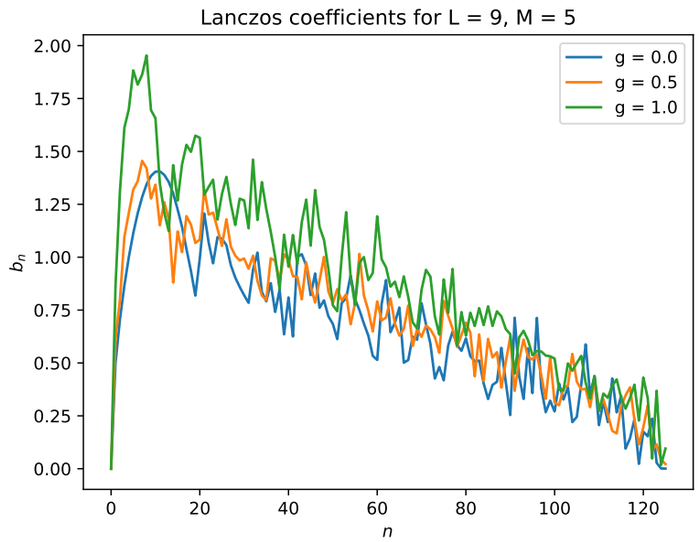


Figure 4.2: Lanczos coefficients in a shorter chain

Unfortunately, due to the computation time scaling and limited computer resources, we were only able to compute the Lanczos coefficients up to chain length of 15 with 8 spin ups. Going any further would have been infeasible due to the way that the time required per computation scales with $\dim(\mathscr{H})$.

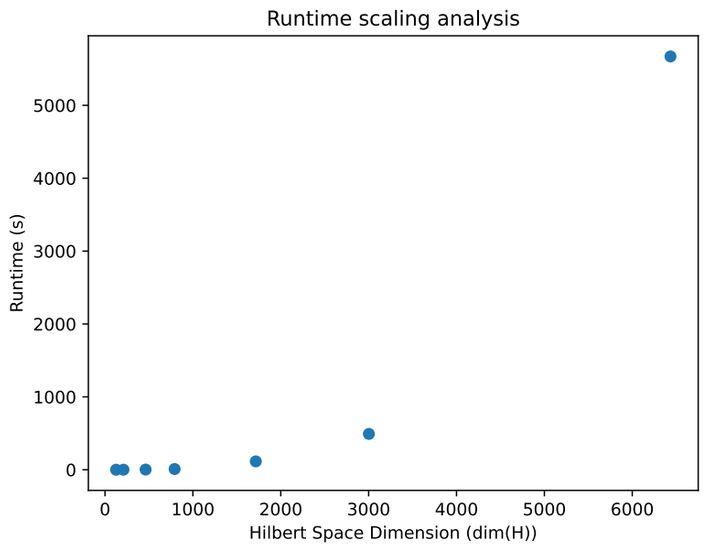


Figure 4.3: Time Scaling of the Lanczos coefficients computation

We see that the runtime scales exponentially. This is likely due to the Graham-Schmidt re-orthogonalisation at each step of the Lanczos algorithm, which is unfortunately needed to keep numerical error from building up.

4.3 Krylov Complexity

Using the Lanczos coefficients, the Krylov Complexity was computed as outlined above in Equation 3.25. The plots of all the complexities computed can be seen below:

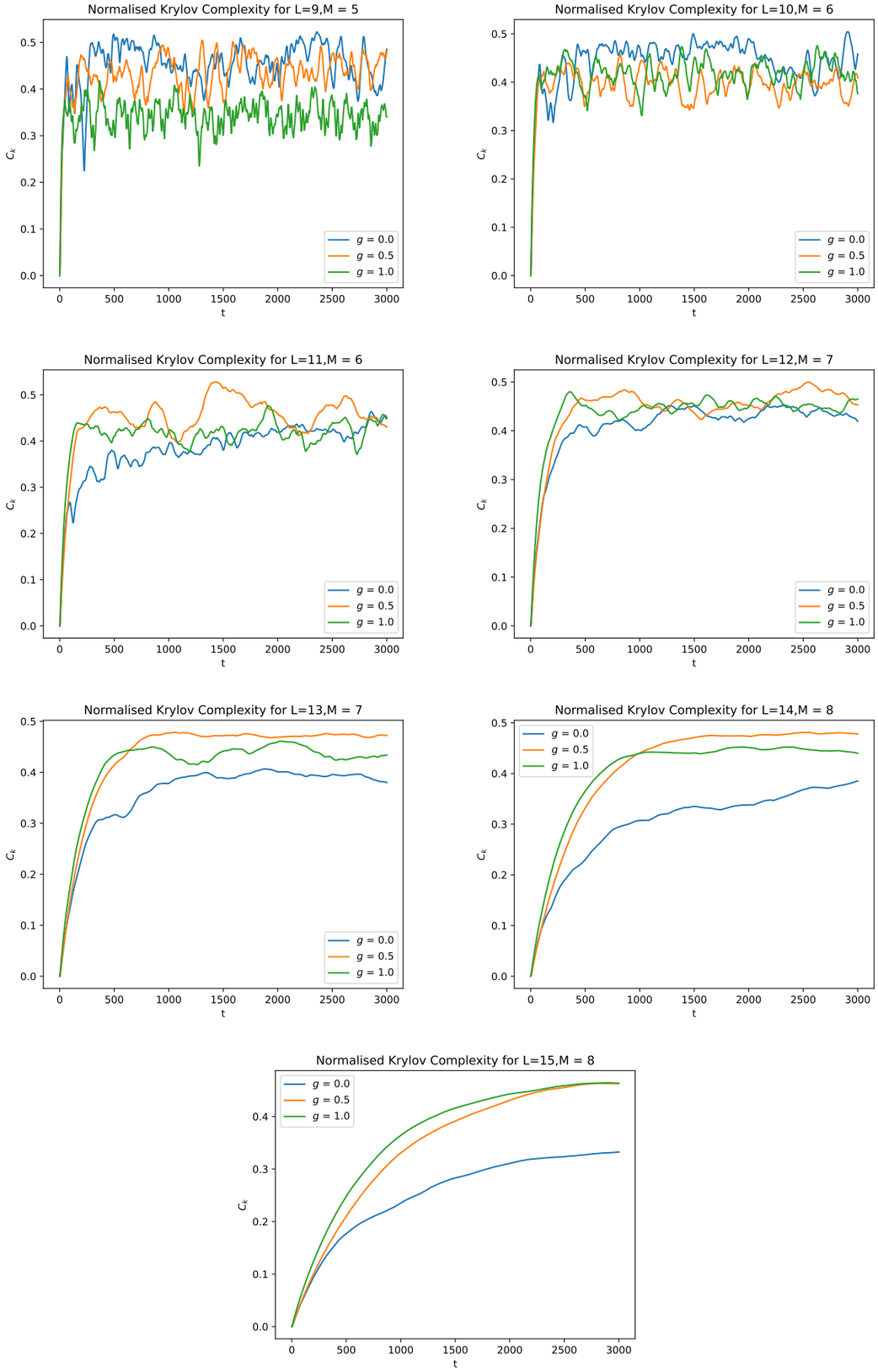


Figure 4.4: Krylov complexity for differing lengths

Note that here the Krylov Complexity has been normalised by simply dividing it by the dimension of the Krylov space. This was done in order to compare the complexities between chains of different lengths.

We see many of the features we expect from Krylov Complexity, such as reaching a plateau after a certain amount of time passed. Shorter chains are much too noisy to really make out a pattern however, for longer chains it is clear that C_K attains a higher plateau value than in the integrable case, as well as attaining it faster. This is expected, since due to longer range interactions, the states are able to "mix" faster.

The difference between integrable and chaotic systems becomes much clearer for longer chains. This supports the hypothesis that the transition from integrable to chaotic becomes more pronounced in the thermodynamic limit.

4.4 Shannon Entropy

We continue to study the transition of the Heisenberg XXZ model with Next to Nearest Neighbour (NNN) interaction in this section, however we now focus on a different quantity, the Shannon entropy in the standard basis. There has been some work on investigating Shannon entropy as a signifier of chaos, we seek to examine this ourselves, verifying our previous findings in the realm of Krylov complexity. In the following, we find the eigenstates of the Hamiltonian and take the Shannon entropy of those states.

The numerical results for Shannon entropy were largely what can be expected from previous work on the topic. Generally, the more "chaotic" the system, the more the Shannon entropy of the eigenstates of the Hamiltonian seem to have a functional relationship with the eigenvalues. It's also relatively clear that as the dimension of the Hilbert space rises, the average Shannon entropy value does too, and it begins to level off.

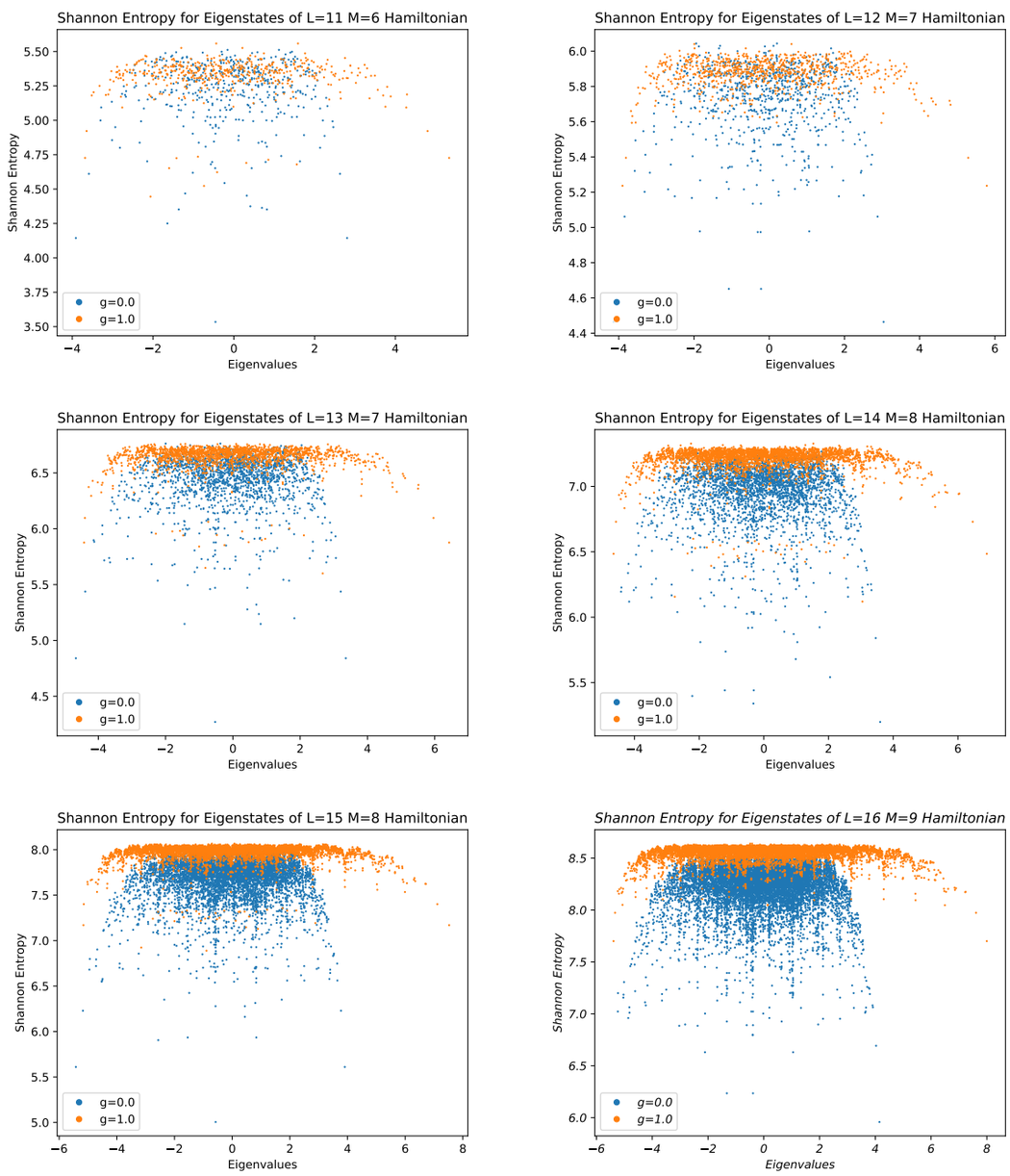


Figure 4.5: Shannon entropy for differing lengths

As the dimension of the Hilbert space grows, the variance in the data steadily decreases and approaches a minimum. For any set of values, as the number of data points in the set increases, the variance of the set approaches the value of the true underlying distribution. The variance of the sample variance decreases at a rate of $\frac{1}{\sqrt{n}}$, where n denotes the sample size of the population. This statistical convergence would occur regardless of the physical nature of the system.

The physical interpretation of this decrease depends on two main scenarios. The first being that the observed entropy distribution is approaching its true underlying distribution, the variance decreasing due to improved sampling. The second understanding is that the true underlying entropy distribution itself is

changing with the Hilbert space dimension. With more data points this could be something to uncover, however due to computational limitations we were unable to gather meaningful results.

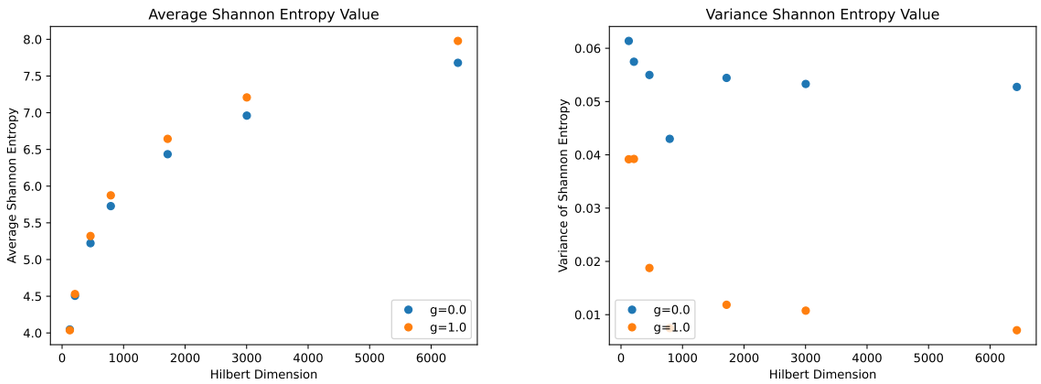


Figure 4.6: Shannon entropy results for different Hilbert dimensions

Our work in this section supports our findings in section 4.3. Generally, the distinction between chaotic and integrable systems become visible and apparent at large Hilbert spaces, in other words, the distinction becomes clearer as we approach the thermodynamic limit. However, there is something to be said of sample size, it's possible that this isn't just a physical result and instead is due to pure lack of data. Although our findings validate the hypothesis that the chaotic behaviour of the system becomes more distinct approaching the thermodynamic limit, more investigation should be done at longer length chains.

5 Conclusion

This project successfully verified that Krylov complexity functions as a good diagnostic tool for chaotic systems. By examining the XXZ chain with Next-to-Nearest-Neighbour (NNN) interaction, we uncovered some notable results. Primarily, we found Krylov complexity distinguishes between chaotic and integrable systems, reaching higher plateaus for the chaotic case. Chaotic systems were also shown to thermalise faster, reaching their plateau values quicker than the integrable case. This indicates enhanced state mixing due to longer range interactions. It's apparent that the distinction between integrable and chaotic behaviour becomes more pronounced as the size of the Hilbert space grows. Similarly, the Shannon entropy of eigenstates shows stronger functional dependence on eigenvalues as we approach the thermodynamic limit.

Krylov complexity offers a dynamical view of quantum chaos, giving us an insight into information propagation and state spreading over time in a system. Our results support the hypothesis that as one approaches the thermodynamic limit the integrable to chaotic transtion becomes progressively more distinct and well defined.

The findings of this project would be greatly supported by further investigation into longer length chains and different NNN coupling coefficients. Unfortunately, due to computational restraints, we were limited to rather small Hilbert dimensions in this project. More investigation into intermediate NNN couplings ($0 < g < 1$) could reveal useful information about the integrable to chaotic transition. Extension to operator complexity, although computationally more intensive, would also provide additional information about the behaviour of systems as they approach the thermodynamic limit.

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